Measuring Equity in Urban Mobility Infrastructure: A Composite-Index and Interactive-Dashboard Approach

Initial Draft — Chapters 1–3 EMGT 7945 · Engineering Management Capstone Aashritha Kanagala · Northeastern University · Summer 2026 Submitted: June 15, 2026

> *Note on draft status.* This draft establishes the problem, situates it in the literature, and specifies the methodology in full. The data pipeline, Mobility Equity Score (MES) computation, and interactive dashboard are implemented and run end-to-end for the three Phase-1 cities (Chicago, Houston, Seattle) on **measured open data**: walkability (EPA Smart Location Database), bicycle (OpenStreetMap), EV charging (OpenStreetMap `amenity=charging_station`), and demographics (Census ACS 5-year 2019–2023) for all three cities, plus transit (GTFS) for Chicago (CTA) and Seattle (King County Metro). The single remaining placeholder is the **Houston transit** layer, for which no key-free GTFS endpoint is currently available (RideMetro requires API registration); it is clearly labelled and excluded from interpretation. Two data caveats are carried forward to Chapter 5: the EV layer uses community-maintained OSM charging points (a lower-bound proxy relative to the authoritative DOE/NREL AFDC inventory, which was unreachable on the development network), and the measures are supply-side throughout. Chapter 4 (Results) is forthcoming; no placeholder values are interpreted as findings in this draft.

Chapter 1 — Introduction

1.1 Background and Motivation

Mobility is a precondition for opportunity. Access to jobs, education, health care, and social participation is mediated by the transportation options available where a person lives. Yet the infrastructure that enables mobility — frequent public transit, walkable streets, connected bicycle networks, and, increasingly, electric-vehicle (EV) charging — is not distributed evenly across urban space. Decades of transportation and land-use research have documented that low-income neighborhoods and communities of color frequently experience a "mobility burden": longer trips, fewer modal choices, and higher exposure to the costs of car dependence (Litman, 2023; Karner et al., 2020). As cities pursue decarbonization and "20-minute neighborhood" goals, the question of *who* benefits from new mobility investment — and who is left behind — has moved to the center of transportation policy.

The challenge for planners and researchers is that mobility equity is multidimensional and spatially heterogeneous. A neighborhood may enjoy excellent bus service but lack safe cycling infrastructure; another may be highly walkable yet far from any fast EV charger as the vehicle fleet electrifies. Existing public tools tend to measure one dimension at a time. The U.S. EPA's Smart Location Database characterizes the built environment and walkability; the U.S. Department of Energy's Alternative Fuels Data Center (AFDC) inventories charging stations; transit agencies publish service data through the General Transit Feed Specification (GTFS). No widely-used, reproducible tool combines these supply-side layers into a single, neighborhood-level measure of mobility access that can be compared across cities and cross-referenced against demographic disadvantage.

1.2 Problem Statement

This project addresses that gap by constructing a **Mobility Equity Score (MES)** — a composite 0–100 index computed at the U.S. Census tract level that integrates four mobility dimensions (transit access, walkability, bicycle infrastructure, and EV-charging access) and by delivering an **interactive dashboard** that lets analysts and policymakers explore the score, identify "mobility deserts," and examine how mobility access correlates with neighborhood demographics. The work is explicitly *supply-side*: it measures the infrastructure made available to a neighborhood, not the travel behavior or realized accessibility of its residents. This scope is a deliberate, stated limitation (§5) and shapes the interpretation of every result.

1.3 Research Questions

1. **Measurement.** Can heterogeneous, publicly-available mobility datasets be integrated into a single, transparent, tract-level composite index that is reproducible and comparable across cities?
2. **Distribution.** How is mobility infrastructure spatially distributed within each study city, and where are the "mobility deserts" (tracts in the bottom quartile of the MES) and "compounded deserts" (bottom quartile on two or more dimensions)?
3. **Equity.** Is the MES associated with neighborhood demographic characteristics — median household income, racial composition, vehicle access, and tenure — and in what direction and magnitude?
4. **Robustness.** Are the conclusions sensitive to methodological choices, specifically the weighting of sub-indices and the percentile threshold used to define a mobility desert?

1.4 Objectives

- Build a modular, documented Python pipeline that ingests eight public data sources and produces tract-level sub-indices for four mobility dimensions.
- Define and compute the MES under a transparent normalization and weighting scheme, with mobility-desert and compounded-desert classifications.
- Quantify the relationship between the MES and demographic disadvantage using correlation analysis, and characterize spatial clustering using spatial autocorrelation and unsupervised clustering.
- Test robustness through weighting-scheme and threshold sensitivity analyses.
- Deliver an interactive Plotly Dash dashboard for exploration and comparison, plus static figures and tables suitable for an academic report and policy brief.

1.5 Scope and Significance

The Phase-1 study covers three structurally different U.S. cities — Chicago (mature, high transit), Houston (auto-oriented, sprawling), and Seattle (progressive transit and EV investment) — chosen to stress-test the method across mobility regimes. The contribution is threefold: (i) a *reproducible integration* of mobility datasets that are usually analyzed in isolation; (ii) a *composite equity metric* with explicit, testable methodological choices; and (iii) a *decision-support dashboard* that translates the analysis for non-technical stakeholders. The significance is practical — the tool can help agencies prioritize investment toward under-served neighborhoods — and methodological, in demonstrating how open data and modern geospatial tooling can operationalize "mobility justice" concepts at scale.

1.6 Organization of the Report

Chapter 2 reviews the literature on transportation equity, composite indicator construction, and the four mobility dimensions. Chapter 3 specifies the methodology in full: study-area selection, data sources, preprocessing, sub-index computation, normalization and MES construction, weighting and threshold sensitivity, and the statistical and dashboard methods. Chapters 4 and 5 (forthcoming) present results and discussion.

Chapter 2 — Literature Review

2.1 Transportation Equity and Mobility Justice

Transportation equity scholarship distinguishes *horizontal equity* (equal treatment of equals) from *vertical equity* (favoring the disadvantaged), and increasingly frames access to mobility as a matter of justice rather than mere efficiency (Martens, 2017; Litman, 2023). Karner et al. (2020) synthesize the field and argue for distributional analyses that pair infrastructure or service measures with demographic data at fine spatial resolution — precisely the design adopted here. The "mobility justice" literature (Sheller, 2018) further situates transportation access within broader patterns of spatial and social inequality, motivating the cross-referencing of the MES against income, race, vehicle access, and tenure.

A persistent methodological tension is between *accessibility* measures (opportunities reachable within a travel-time budget; Levine et al., 2012) and *supply* measures (infrastructure present in or near a place). Accessibility is theoretically richer but data-intensive and behaviorally contingent; supply measures are transparent and reproducible but silent on whether residents can actually use the infrastructure. This project adopts a supply-side framing for reproducibility and cross-city comparability, while explicitly acknowledging the trade-off (§5).

2.2 Composite Indicators in Public Policy

Composite indicators compress multiple variables into a single, communicable number; the OECD/JRC *Handbook on Constructing Composite Indicators* (Nardo et al., 2008) is the canonical methodological reference and structures the choices made in Chapter 3: framework definition, data selection, treatment of missing data, normalization, weighting and aggregation, and robustness/sensitivity analysis. The handbook stresses that normalization and weighting are value-laden and that any composite must be accompanied by uncertainty and sensitivity analysis — a requirement this study meets through its weighting-scheme and threshold tests (§3.6, §3.7). Well-known applied examples (e.g., the UNDP Human Development Index; the CDC/ATSDR Social Vulnerability Index) demonstrate both the communicative power and the contestability of composites, reinforcing the need for transparency about every modeling decision.

2.3 The Four Mobility Dimensions

Transit access. Service quality is commonly proxied by stop density, frequency (trips per hour), and span of service, computed from GTFS feeds within a walkable buffer of residences (typically 0.25–0.5 mi). GTFS has become the de facto standard for transit supply analysis, and open libraries now make feed-level frequency computation tractable.

Walkability. The EPA Smart Location Database (SLD) provides the National Walkability Index (`NatWalkInd`, 1–20), a block-group measure built from intersection density, transit proximity, and employment/household mix. It is the most widely-used national, consistently-computed walkability measure and is adopted here directly, with areal interpolation to harmonize its 2019-vintage block groups to the study's 2023 tracts (§3.3).

Bicycle infrastructure. OpenStreetMap (OSM), accessed via the OSMnx library (Boeing, 2017), is the standard open source for bicycle-network extraction. Dedicated infrastructure (cycleways, paths, tracks, and marked on-street lanes) can be filtered from the cyclable network and summarized as lane length per unit area, a common supply proxy for bikeability.

EV-charging access. As fleets electrify, equitable access to public charging — especially DC-fast charging for households without home charging — is an emerging equity concern. The DOE/NREL AFDC is the authoritative U.S. inventory of public stations and supports density and proximity measures at the neighborhood scale.

2.4 Spatial Analysis Methods

Because mobility infrastructure is spatially structured, the analysis employs established spatial-statistical tools. Global and local Moran's I (Anselin, 1995), implemented in the PySAL ecosystem (Rey & Anselin, 2007), test whether MES values cluster in space and locate statistically significant low-low "cold spots." Unsupervised K-means clustering with silhouette-based model selection (Rousseeuw, 1987) is used to derive an interpretable typology of neighborhood mobility profiles, complementing the single-number MES with a multidimensional classification.

2.5 Existing Tools and the Gap Addressed

Agencies and researchers can consult the EPA SLD for walkability, the AFDC for chargers, transit agencies' GTFS for service, and bespoke bike-network analyses. National equity screening tools (e.g., the USDOT Equitable Transportation Community Explorer; the CDC Social Vulnerability Index) provide disadvantage indices but are not mobility-infrastructure composites at tract level across modes. The gap this project fills is an *integrated, reproducible, cross-city, multi-modal* mobility-supply composite with an exploratory dashboard — bringing the four dimensions into a single comparable frame and pairing them with demographic context.

Chapter 3 — Methodology

3.1 Study Area Selection and City FIPS Codes

Three cities were selected to span distinct mobility regimes (Table 3.1). The analytical unit is the 2023 Census tract; counties are identified by FIPS code for data retrieval.

Table 3.1 — Study cities.

City	State (FIPS)	County (FIPS)	Tracts	Rationale
Chicago	Illinois (17)	Cook (031)	1,332	Mature, high-frequency transit; diverse demographics
Houston	Texas (48)	Harris (201)	1,115	Auto-oriented, sprawling; large low-income population
Seattle	Washington (53)	King (033)	495	Progressive transit and EV investment; equity focus

Phase 2 will extend the method to Phoenix (AZ), Atlanta (GA), and Detroit (MI) after pipeline validation.

3.2 Data Collection Pipeline

Eight public sources feed the pipeline (Table 3.2). Each is retrieved by a dedicated, independently-runnable Python module under `src/data_collection/`.

Table 3.2 — Data sources and access.

#	Layer	Source	Endpoint / file	Key
1	Tract boundaries (2023)	Census TIGER/Line	`…/TIGER2023/TRACT/tl_2023_{ss}_tract.zip`	none

2	Block groups (2023, 2019)	Census TIGER/Line	`…/TIGER{2023,2019}/BG/tl_{yyyy}_{ss}_bg.zip`	none
3	Demographics (ACS 5-yr 2019–2023)	Census API	`api.census.gov/data/2023/acs/acs5`	Census
4	Walkability	EPA Smart Location Database v3	`edg.epa.gov/…/EPA_SmartLocationDatabase_V3_Jan`	none
5	Bicycle network	OpenStreetMap (OSMnx)	Overpass via `graph_from_place(..., 'bike')`	none
6	EV charging	DOE/NREL AFDC	`developer.nrel.gov/api/alt-fuel-stations/v1.js`	NREL
7	Transit service	GTFS (transit.land / agency)	`transit.land/api/v2/rest/feeds`; agency zips	transit.land (opt.)
8	(derived) Tract centroids & areas	computed	—	none

ACS variables retrieved: median household income (`B19013_001E`); total population and White-alone-non-Hispanic (`B03002_001E`, `B03002_003E`) for percent non-white; zero-vehicle owner/renter households (`B25044_003E`, `B25044_010E`); renter-occupied and total occupied units (`B25003_003E`, `B25003_001E`) for percent renter.

3.3 Data Preprocessing

All layers are reprojected to a common geographic CRS (EPSG:4326) for storage and mapping and to USA Contiguous Albers Equal Area (EPSG:5070) for any length/area/distance computation, so metrics are in true metres and not degrees. GEOIDs are coerced to zero-padded strings (11 digits for tracts, 12 for block groups) to guarantee exact joins.

A key preprocessing problem is **geographic vintage mismatch**. The EPA SLD v3 is published on 2019-vintage (2010-decade) block-group boundaries, whereas the study uses 2023 (2020-decade) tracts to align with the 2019–2023 ACS. A naive GEOID-prefix join loses tracts wherever boundaries were redrawn — empirically, about 57% of Houston's tracts. The pipeline therefore performs **areal interpolation**: `NatWalkInd` is joined to 2019 block-group polygons (a 100% GEOID match against the SLD) and then area-weighted onto 2023 tracts via a geometric overlay, with each tract's value the area-weighted mean of its overlapping block-group pieces. Because `NatWalkInd` is an intensive index (not a count), area-weighted averaging is the correct interpolation.

Missing values are preserved rather than imputed so that data gaps remain visible; normalization functions propagate `NaN`, and a constant input maps to the scale midpoint rather than producing undefined values. Tracts with zero land area or no population are guarded against division-by-zero in density metrics.

3.4 Sub-Index Computation

Each sub-index is computed at tract level and then normalized (§3.5).

Transit (`transit_index.py`). From GTFS `stops` and `stop_times`, AM-peak (07:00–09:00) departures per stop are converted to trips/hour, and the daily service span (distinct hours with ≥1 departure) is computed. Stops are spatially joined to a 0.5-mile (804.7 m) buffer of each tract centroid, yielding stop density (stops/km²), mean AM-peak frequency, and service span. The three components are normalized and averaged.

Walkability (`walk_index.py`). Area-weighted `NatWalkInd` per tract (§3.3), normalized to 0–100.

Bicycle (`bike_index.py`). From the OSM cyclable network, edges are retained as dedicated infrastructure when `highway ∈ {cycleway, path, track}` or a positive `cycleway` tag is present (e.g., `lane`, `track`, `shared_lane`). Edges are clipped to tracts by overlay; bike-lane kilometres per km² of tract land area form the raw metric, normalized to 0–100.

EV charging (`ev_index.py`). Public Level-2 and DC-fast stations are spatially joined to tracts. Two metrics are computed: chargers per 1,000 residents, and the distance from each tract centroid to the nearest DC-fast station (inverted so that proximity raises the score). Both are normalized and averaged. The authoritative source is the DOE/NREL AFDC inventory (`fetch_afdc.py`); where that API is unreachable, the pipeline falls back to OpenStreetMap `amenity=charging_station` points (`fetch_osm_ev.py`), a key-free but lower-bound proxy whose completeness limitation is noted in §5.

3.5 Normalization and MES Construction

Each raw sub-index is placed on a common 0–100 scale by **min–max normalization**:

```
x_norm = (x − x_min) / (x_max − x_min) × 100
```

computed within each city so scores are interpretable relative to that city's own distribution. A robust percentile-rank variant is retained for sensitivity checks. The composite is a weighted sum of the four normalized sub-indices:

```
MES = w_transit·transit_norm + w_walk·walk_norm + w_bike·bike_norm + w_ev·ev_norm,   Σw = 1
```

The **baseline** uses equal weights (0.25 each), the most defensible default absent strong prior information (Nardo et al., 2008). The MES therefore ranges 0–100, with higher values indicating better mobility-infrastructure access.

3.6 Weighting Methodology and Sensitivity Analysis

Because equal weighting is itself a value choice, three alternatives are computed and compared (`weighting_methods.py`):

- **Entropy weighting** — objective weights derived from each sub-index's dispersion; dimensions that discriminate more between tracts receive more weight (redundancy = 1 – Shannon entropy, normalized to sum to 1).
- **PCA weighting** — weights from the absolute loadings of the first principal component of the four sub-indices.
- **AHP weighting** — expert pairwise-judgement weights (Analytic Hierarchy Process); a provisional vector (transit 0.40, walk 0.30, bike 0.15, EV 0.15) is used pending a structured expert survey.

For each scheme the MES is recomputed and the equity correlations (§3.8) are re-estimated; the analysis reports how much the sign, magnitude, and significance of those correlations move across schemes, establishing whether conclusions are weighting-robust.

3.7 Mobility-Desert Classification

A tract is a **mobility desert** if its MES falls below the city's 25th percentile. A **compounded desert** is a tract in the bottom quartile of two or more individual sub-indices — capturing places disadvantaged on multiple dimensions simultaneously. Because the 25th-percentile cut is arbitrary, a **threshold sensitivity analysis** (`sensitivity_analysis.py`) recomputes the desert set at the 20th, 25th, and 30th percentiles and reports the Jaccard overlap with the baseline set and the demographic profile of flagged tracts at each threshold; high overlap indicates the classification is not an artefact of the chosen cut-off.

3.8 Equity Correlation Analysis

Pearson correlations relate the MES (and each sub-index) to four demographic variables — median income, percent non-white, percent zero-vehicle households, and percent renter (`equity_correlation.py`). For each pair the analysis reports r, the two-sided p-value, the sample size, and the 95% confidence interval from the Fisher z-transformation:

```
z = arctanh(r),  SE = 1/√(n−3),  CI = tanh(z ± 1.96·SE)
```

A negative MES–income or MES–percent-non-white correlation would indicate that more-disadvantaged neighborhoods receive less mobility infrastructure (a vertical- equity concern). All correlations are also rendered as a correlation-matrix heatmap for the report.

3.9 Spatial Autocorrelation (Moran's I)

Global Moran's I (`spatial_autocorrelation.py`) tests whether MES values cluster spatially, using **queen-contiguity** weights, row-standardized, with 999 conditional permutations for a pseudo p-value. Local Moran's I (LISA) classifies each tract as high-high, low-low, high-low, low-high, or non-significant ($p < 0.05$); low-low clusters are mapped as mobility "cold spots."

3.10 Cluster Analysis

K-means clustering (`clustering.py`) on the four standardized sub-indices derives a typology of neighborhood mobility profiles. The number of clusters k is chosen by maximizing the mean silhouette score over $k \in \{2,\ldots,7\}$, with the inertia (elbow) curve reported for validation. Each cluster is given an interpretable label from its centroid profile (e.g., "uniformly under-served," "well-served / multimodal," or "transit-rich, bike-poor") for the Chapter 4 typology table.

3.11 Dashboard Development

The interactive dashboard (`src/dashboard/`, Plotly Dash + Dash Bootstrap Components) follows a modular layout/callbacks/utilities architecture. It provides: a tract-level choropleth of the MES or any sub-index (reversed yellow-orange-red scale, darkest = worst) with desert filters; a click-through side panel showing a tract's MES, city percentile, a sub-index bar chart versus the city median, an auto-generated plain-language interpretation, and a demographic summary; a city-comparison equity-gap chart (median MES of the poorest versus richest income quintile); a ranked table of the most disadvantaged tracts; and CSV/PNG export. Each MES file records the measurement status of every layer so placeholder data is never displayed as measured.

3.12 Reproducibility and Tooling

The project targets Python 3.11 with a pinned `requirements.txt`. Every stage is an independently-runnable module, raw downloads are cached, and a `pytest` suite covers normalization, MES construction and desert flagging, and the correlation statistics. Code, data-processing scripts, and documentation are version- controlled, with raw and large artefacts gitignored.

References (selected; to be completed in final draft)

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